

10:00 – 23:59hrs

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EXAMS OFFICE
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University of the Witwatersrand, Johannesburg

Course or topic No(s)

COMS4060A/7056A

Course or topic name(s)
Paper Number & title

Intro to Data Visualisation and Exploration

Examination to be held
during the month(s) of

July, 2022

Year of study

Degrees/Diplomas for
which this course is
prescribed

Faculties presenting can-
didates

Science

Internal examiner(s)

Dr Timothy Bristow

External examiner(s)

Dr Paresh Malalur

Special materials

Dataset Provided on Moodle

Time allowance

108 Hours

Instructions to candidates

163 Marks available. 150 marks = 100%.
Answer all questions. This is an **open-book** exam:
you may consult any material. Do **not** copy code
from any external sources. This exam consists of 16
pages and 3 questions. Submit a Jupyter notebook
or PDF on Moodle by 23:59 05/07/2022.

Introduction to Data Visualisation and Exploration

Datasets and Code

You can find all of the *required* datasets available for download on Moodle in `datasets.zip`.

You can find and use code from the Jupyter notebooks provided and linked throughout the course for performing many parts (but not all) of this exam.

Question 1 Medical Insurance [40 Marks]

Dataset Overview

You are provided with a dataset, `insurance.csv`, containing medical insurance claims for anonymous clients in the US. The dataset contains some information on the clients which might have some correlation with the total amount of all claims (charges). You can assume that all clients in this dataset have the same exposure: i.e., all claims for all users are taken over the same period of time. Perform exploratory data analysis and provide visualisations of this dataset to answer questions based on the data. Ostensibly, the purpose of this dataset is to find correlations between total medical costs and the other factors.

Insurance companies make money by collecting more in yearly premiums than they spend on medical care for its beneficiaries. As a result, insurers invest time and money to develop models that accurately forecast medical expenses. These are difficult to estimate because the most costly conditions are rare and seemingly random. Still, some conditions are more prevalent for certain segments of the population, eg. lung cancer is more likely among smokers than non-smokers, and heart disease may be more likely among the obese.

The goal of this analysis is to use patient data to estimate the average medical care expenses for such population segments. These estimates could be used to create actuarial tables which set the price of yearly premiums higher or lower depending on the expected treatment costs. Note: South Africa has standardised medical aid rates, which is different from the US and many other countries, where medical insurance is priced according to the person (like life insurance in SA).

The insurance data contains the following:

- `age`: age of primary beneficiary.
- `sex`: gender of the primary beneficiary: female, male.
- `bmi`: body mass index. Gives a relationship between weight and height, providing an objective index of body weight (kg/m^2) using the ratio of height to weight, ideally 18.5 to 24.9. There is a BMI chart in the `datasets.zip` file.

- `children`: number of children covered by the insurance plan (or how many dependents there are).
- `smoker`: if the primary beneficiary is a smoker.
- `region`: the primary beneficiary's residential area in the US: northeast, southeast, southwest, northwest.
- `charges`: total medical costs (claims) billed to the insurance company for this period.
- `premium`: monthly premium paid by the client for this insurance plan.

1.1 Describing and Cleaning the Data

1. Describe the dataset: number of entries, range/mean of the variables, number of unique regions, for example. [1]
2. Identify missing values and duplicates. Considering that `charges` is the target variable, decide if rows with this value missing should be imputed or dropped. Perform this cleaning on the dataset, and use this going forward. [3]
3. Convert variables to relevant datatypes: categorical/one-hot encoding, etc. [2]

1.2 Exploring the Data

For the below, show **plots** where required and **always interpret** the data.

4. Plot the `charges` and comment on the distribution and any outliers you can identify. [2]
5. Is there a correlation between `charges` and the other variables? Provide a correlation matrix to show this. [3]
6. Provide a graph showing the number of smokers and non-smokers. Then plot the `charges` by smoking status. There appear to be tails in the charges for non-smokers. What would explain this? [3]
7. How is the dataset divided by `sex`? Is there a difference in `charges`? [3]
8. Look at the relationship between `children` and `charges` and comment on the results. [2]
9. Plot the relationship between `children` and `age` and comment on the results. [2]
10. Plot the `age` distribution. What do you find unusual about this distribution? [2]
11. How do insurance charges change by age? Apart from the expectation that older clients are less healthy and have larger claims, what other factors might influence this? [3]
12. Is there a relationship between smoking and age? [2]

13. Plot the BMI distribution. What can you deduce from this distribution? What is the difference in charges for clients that are deemed obese (and greater) vs non-obese (and lower)? [4]
14. Is there a relationship between the `region` and `age/charges/smoking`? One region has higher charges than the others. Which region is this? What might impact this? [4]

1.3 Identifying the Highest Risks

Based on the above findings, there is an indication of which variables might be the most informative when predicting the cost of claims.

15. The correlation matrix should indicate that `smoker` is strongly correlated with charges. Referring to the plots from earlier showing the charges for smokers, there should be a clear separation in higher and lower charges. What is causing this? What variable(s), when combined with `smoker=True` indicates a much higher claim amount? Provide plots to motivate your answer. [4]

From this, it should be clear that there are specific risk factors that increase the overall medical costs. In South Africa, medical insurers cannot discriminate in any way when calculating a monthly premium for someone - everyone pays exactly the same amount. When actuaries are pricing the insurance premiums, they need to make sure the premium will cover costs for the entire population of clients. Having a healthier population benefits everyone.

Question 2 Human Activity Recognition [50 Marks]

Dataset Overview

Activity recognition using smartphones (or smart devices more generally) has many use cases and potential. Examples include sports tracking, or monitoring elderly or less-abled people. This is a difficult problem to solve for several reasons: people have different movements, devices have different components and capabilities (different phones, watches, etc), the implementation might change (Android vs iOS), and the design and implementation of algorithms that can identify the activities.

A well-known dataset (which was discussed in the lectures) is the Human Activity Recognition dataset from D. Anguita et. al. See here for details <https://archive.ics.uci.edu/ml/datasets/Human+Activity+Recognition+Using+Smartphones>.

Description of the dataset from the authors:

The experiments have been carried out with a group of 30 volunteers within an age bracket of 19-48 years. Each person performed six activities (WALKING, WALKING_UPSTAIRS, WALKING_DOWNSTAIRS, SITTING, STANDING, LAYING) wearing a smartphone (Samsung Galaxy S II) on the waist. Using its embedded accelerometer and gyroscope, we captured 3-axial

linear acceleration and 3-axial angular velocity at a constant rate of 50Hz.

The sensor signals (accelerometer and gyroscope) were pre-processed by applying noise filters and then sampled in fixed-width sliding windows of 2.56 sec and 50% overlap (128 readings/window). The sensor acceleration signal, which has gravitational and body motion components, was separated using a Butterworth low-pass filter into body acceleration and gravity. The gravitational force is assumed to have only low frequency components, therefore a filter with 0.3 Hz cutoff frequency was used. From each window, a vector of features was obtained by calculating variables from the time and frequency domain.

A complete description of the data is given in the attached file: `UCI HAR Dataset.names`.

For each record in the dataset the following measurements are provided:

- Triaxial acceleration from the accelerometer (total acceleration) and the estimated body acceleration.
- Triaxial Angular velocity from the gyroscope.
- A 561-feature vector with time and frequency domain variables (many of which are derived from the accelerometer and gyroscope readings, see below).
- Its activity label.
- An identifier of the subject who carried out the experiment.
- Note that each window is 1.28 seconds long.

The following terms might be useful in deciphering the data (these are included for most of the signals):

- `mean()`: Mean value
- `std()`: Standard deviation
- `mad()`: Median absolute deviation
- `max()`: Largest value in array
- `min()`: Smallest value in array
- `sma()`: Signal magnitude area
- `energy()`: Energy measure. Sum of the squares divided by the number of values.
- `iqr()`: Interquartile range
- `entropy()`: Signal entropy
- `arCoeff()`: Autoregression coefficients with Burg order equal to 4
- `correlation()`: correlation coefficient between two signals

- `maxInds()`: index of the frequency component with largest magnitude
- `meanFreq()`: Weighted average of the frequency components to obtain a mean frequency
- `skewness()`: skewness of the frequency domain signal
- `kurtosis()`: kurtosis of the frequency domain signal
- `bandsEnergy()`: Energy of a frequency interval within the 64 bins of the FFT of each window.
- `angle()`: Angle between two vectors.

2.1 Exploration

When working with this dataset, the typical goal is to classify activity into one of the labels: walking, lying down, etc. Here, the goal is exploratory data analysis, which is done *before* the classification step. Perform your exploration on the *training* dataset.

1. Describe the dataset: number of entries, range/mean of the variables. [1]
2. How many of each class? How many for each user? Provide basic visualisations for this. [2]

Broadly, these activities can be divided into dynamic and static movements (walking vs sitting or lying down, for example).

3. Identify and plot at least two features that show a clear static/dynamic separation. Your plots must convey this information clearly. [3]
4. Identify a simple rule (if/else, basically) on this feature that can be used to separate dynamic from static movements. [1]
5. Using this rule only, what would the accuracy be if trying to predict static vs dynamic movement on the test dataset?

Intuitively, one would expect that lying down would have minimal movement, but have an identifiable angle between the phone sensor axes and the direction of gravity.

6. Identify and plot a feature that exploits this, showing class separation. [2]
7. Give a simple rule that effectively separates LAYING from all other classes. [1]
8. Using this rule only, what would the accuracy be if trying to identify LAYING on the test dataset? [2]

Another question we might ask is if we can identify individual *users* based on data for specific activities.

9. Perform PCA on this dataset. What is the explained variance when performing PCA with the following number of components: 2, 5, 10? [5]
10. Which variables contribute the most to the first two principal components? What do you notice about the types of variables? [4]

11. Now, perform the above PCA separately on static and dynamic movements. Does the number of components explain as much of the variance for each? Which variables contribute the most to the first two principal components? [4]

2.2 Baseline models

In the above question, you should have identified that without applying any sophisticated modeling you are already able to get insights and draw valuable information from the data. We will now start looking at some baseline models to get more out of the data.

Define three baseline models (with reasonable parameters) for this section (all of these are in sklearn, and you have seen RFC and logistic regression in the example notebooks during class):

- Random Forest Classifier
- Logistic Regression

When setting up the models, always set `random_state=42`. [1]

Use F_1 score to measure the performance of your classification. Remember, $F_1 = 2 \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$, where $\text{Recall} = \text{True Positives} / (\text{True Positives} + \text{False Negatives})$ and $\text{Precision} = \text{True Positives} / (\text{True Positives} + \text{False Positives})$. However, you can use the built-in functions in `sklearn` if you are using Python.

Do not forget to test on the *test* dataset! Interpret and comment on your findings.

1. Use the three baseline models with all 561 features to predict the activity type. Record the F_1 score and the time taken to train the model. [4]
2. Use the three baseline models with all 561 features to predict if the activity is static or dynamic. Record the F_1 score and the time taken to train the model. How does this compare with your simple decision rule identified earlier? [4]

2.3 Feature Selection

The dataset has 561 features, which likely have some correlations, and not all will be useful (maybe even harmful). Having a large number of features is going to affect the training time of the model too.

1. Perform mutual information feature selection (`mutual_info_selection` in sklearn) to select the top 5, 10, 50, 100 features for identifying activity type. Using the baseline models, investigate how the reduction in features affects the training time and the F_1 score (you need to run each model for all feature sets). Comment on your results. [4]

2.4 Feature Extraction

In the previous section you took the existing features and found subsets of these which could be used for training. Another approach is to either engineer or extract features from the data. Perform the following dimensionality reduction techniques (you have code to perform this from the lectures):

- PCA
- ISOMAP
- UMAP

You need to identify suitable parameters for each of the models: number of components, number of neighbours, etc.

1. Show a 2D plot of the embeddings/principal components, and comment on the results. [4]
2. Train the baseline models on these embeddings (your features are the output from PCA/ISOMAP/UMAP). Record the training time and the F_1 score. [4]
3. Discuss your findings on the trade-off between training time, performance, and interpretability. Compare the baseline models with no feature reduction, when using feature selection, and using the feature extraction methods here. [4]

Question 3 Indian Premier League [73 Marks]

The Indian Premier League is an annual T20 club cricket tournament that (normally) takes place in India. The teams are based in several cities in India, but consist of players from around the world.

From the IPL Wikipedia page:

The IPL is the most-attended cricket league in the world and in 2014 was ranked sixth by average attendance among all sports leagues. The 2020 IPL season set a massive viewership record with 31.57 million average impressions and with an overall consumption increase of 23 per cent from the 2019 season.

Most of the teams are valued at over a billion dollars (as of 2022), and they are allowed to spend up to \$ 11 million on player salaries. There is an annual auction where the club franchises bid for players, similar to many other club sports. Individual players can fetch upwards of \$ 1 million. There is significant money involved in the IPL, and choosing or finding the best players is paramount to success. A team having the top batsman in the world is more likely to pull larger crowds, and, hopefully, win more games. You can see historical auctions here: <https://www.iplt20.com/auction>.

Franchises need to keep track of which players are performing well, and whether they are getting a return on their investment. Similarly, they might want to poach players from other teams in the next year. To do this they need data. There is a significant amount of data generated per game (if you are a cricket fan, you will know how much the statisticians love the sport too) which can give some insight to the performance of players and teams.

3.5 Cricket

For those that are not familiar with cricket, a short 4 minute video has been included with your datasets which gives a brief explanation. Otherwise, here are some additional resources:

- <https://howtheyplay.com/team-sports/cricketfordummies>
- <https://en.wikipedia.org/wiki/Cricket>
- <https://www.espncricinfo.com/ci/content/page/429548.html>
- <https://www.usacricket.org/what-is-cricket/>
- When you have finished this exam, you can watch this video about how cricket is **not** played: <https://www.youtube.com/watch?v=KQ3mdrQeTHE>

The cricinfo description is perhaps less useful:

Cricket is a game played between two teams, generally of 11 members each. In essence, it is single combat, in which an individual batsman does battle against an individual bowler, who has helpers known as fielders. The bowler propels the ball with a straight arm from one end of the 22-yard pitch in an attempt to dismiss the batsman by hitting a target known as the wicket at the other end, or by causing the batsman to hit the ball into the air into a fielders grasp, or by inducing one of a number of other indiscretions. The batsman attempts to defend the wicket with the bat and to score runs - the currency of the game - by striking the ball to the field boundary, or far enough from the fielders to allow the batsman to run to the other end of the pitch before the ball can be returned. At least two bowlers must take turns, from alternating ends; also, there are always two batsmen on the field, each to take a turn as required. When all but one of the batting team have been dismissed - or after an agreed period - the teams roles are reversed. After all the players required to bat on both sides have done so either once or twice (which can take from a few hours to five days) the total number of runs accumulated determines the winner. But sometimes there isn't one.

Tim's description: There is an oval field. In the middle, there is a rectangular pitch, 22 yards long. On either end, there are 3 sticks in the ground (called wickets). This is the battlefield. At any time, there are two batsman - one on either side of this pitch. There are 11 fielders spread around the oval field. One of these is called the bowler. From one set of wickets, they bowl the ball (which is a weird kind of throwing action) towards the batsman on the other side. The batsman uses a cricket bat to try prevent the ball from hitting three wooden sticks, called wickets. If the batman fails to prevent the ball from hitting the wickets, they are out and must leave the field. Similarly, if they hit the ball and it goes up in the air and a fielder catches it, they are out. If the ball was going to hit the wickets, but they put their body in the way, they are out leg before wicket. Batsmen also need to score runs. Runs are scored by the one batsman hitting the ball, and then both batsmen running to the other side of the pitch. They can also hit the ball out of the oval field to get 4 runs (if it bounces), or 6 runs (no bounces). Bowlers take turns, bowling 6 deliveries at a time. These 6 deliveries are called an over. IPL games consist

of 20 overs for each team batting. But really, the best idea is to watch that video that is included with the exam.

Some additional terminology:

- *Delivery*: a single ball bowled (if you liken it to baseball, it's one throw by the pitcher). A delivery is legal if it is not a wide or no-ball (see *Extras* below).
- *Over*: a collection of six *legal* deliveries. Wides and no-balls do not count towards the six ball quota, and need to be bowled again.
- *Extras*:
 - *Wide*: If a bowler has a poor delivery that is too far away from the batsman, a wide is called. A wide gives the batting *team* an extra run (the run does **not** go to the batsman), and the run is counted against the bowler.
 - *No-ball*: In this case, 1 run goes to the batting team, and any additional runs scored go to the batsman. These extras count against the bowler. A player cannot be bowled, caught, lbw, or stumped on a no-ball.
 - *Penalty*: Penalty runs can be given to the batting team, but do not count against the bowler.
 - *(Leg-)byes*: Leg-byes and byes are runs taken by the batsmen where they were not able to hit the ball at all with their bat. These runs go towards the team score, but are not counted towards the batsman's score or the against the bowler.
- *(Non-)Striker*: This is the batsman that is (not) facing the bowler.
- *Toss*: Before the game, the two captains toss a coin to decide who gets to choose if they want to bat or field first. This is down to luck, but can play a big part in the outcome of the game.
- *Powerplay*: There are strict rules about where the fielders can stand. During a powerplay, these restrictions change, and more of the fielders have to stand closer to the batsman. This entices the batsman to try and hit the ball in the air and get boundaries - making it more exciting for the crowd. During a powerplay you would expect runs to be scored at a faster rate, and more wickets to fall.
- *Run rate*: This is the number of runs being scored by the batting team per over. Eg. 100 runs scored in 10 overs means a run rate of 10 per over.
- *Wickets*: The three wooden stumps are called the wickets. Confusingly, if a batsman gets dismissed, this dismissal is also referred to as a wicket. And to make it even more confusing, sometimes the pitch itself is called the wicket! Typically, if talking about a bowler taking wickets, it is referring to the number of batters that they have dismissed.
- *Dismissals*: A batsman can be dismissed or go out in the following ways:
 - *Bowled*: the bowler delivers the ball, the batsman misses it, and the ball hits the wooden wickets,

- leg before wicket (lbw): The batsman misses the ball and it hits their body, but it would have hit the wickets otherwise,
- stumped: the batsman steps towards the bowler (past a painted line called the crease) and tries to hit the ball, misses, and the fielder behind them (wicket keeper) catches the ball and hits the wickets,
- caught: the batsman hits the ball in the air, and it is caught by a fielder,
- run out: when the two batsmen are running to opposite sides of the pitch to score runs, a fielder can try hit the wickets with the ball before they batsmen get to the other side,
- caught and bowled: the batsman hits the ball back to the bowler in the air and they are caught,
- retired hurt/out: a batsman leaves the field because they are injured,
- hit wicket: a batsman accidentally hits the wickets with their own bat,
- obstructing the field: getting in the way of a fielder

Understanding the dataset

You are given data from all IPL matches since 2009. This includes ball-by-ball information. The data is split into two files: a file containing match summary information, and another containing detailed information on all deliveries in the match. You are essentially given the raw data - every delivery of each match, and you need to reconstruct what happened in the game.

The match summary file contains the following fields:

- `match_id`: a unique identifier given to each match.
- `city`: the city where the game was played.
- `venue`: the stadium (ground) where the game was played. Some cities have multiple grounds.
- `match_date`: date on which the game took place.
- `umpire1/2`: names of the umpires for the game.
- `season`: which season this game took place in (ie which year).
- `team1/2`: names of the two teams.
- `toss_winner`: name of the team that won the toss.
- `toss_decision`: if the team that won the toss decided to bat or field.
- `winner`: the winner of the game. Note that in cricket, you can have a no-result or a tie.
- `mom`: the player of the match.
- `inning1_bat`: which team batted first.
- `inning2_bat`: which team batted second.

The delivery file contains all deliveries for all matches. It has, line by line, the match id and the delivery number (in terms of over and ball number in that over):

- `match_id`: unique identifier for the match, which can be used to join up to the match dataset.
- `over`: the over number in which this delivery took place.
- `ball_num`: the delivery number in this over.
- `inning`: if this was the first or second innings of the match.
- `is_pp`: if this was during a powerplay.
- `runs_total`: how many runs were scored for the batting team, in total.
- `runs_extras`: how many of these runs were due to extras.
- `extras_bowler`: how many of the extra runs count against the bowler.
- `extras_team`: how many of the extra runs count against the team.
- `runs_batter`: how many runs went to the batsman on strike.
- `striker`: the name of the batsman that is on strike (ie, who the bowler is bowling to).
- `non_striker`: the name of the second batsman.
- `bowler`: the name of the bowler.
- `how_out`: if a player was dismissed, this will be one of the dismissal types, otherwise it will be none.
- `player_dismissed`: if a player was dismissed, this will tell you the name of the player dismissed - either the striker or non-striker.

3.6 Match Summary Exploration

Before looking at the delivery data, you can get some information from the match summaries. Looking at the teams in the tournaments:

1. How many unique teams are there? (Note, not all teams played every season). You might have to clean up the names of the teams to get the correct answer. [1]
2. Produce a plot showing the win percentage for all teams (bar plot). [2]
3. Plot the win percentage *over time* (ie, how it changes per season) for each team. Comment on these results and if teams are staying the same, improving, or getting worse. [3]
4. Identify the winner of each tournament. You can come up with your own method, and check it against what you find online, but you must explain how you got to this answer. [2]

The toss can influence the game and the tactics used. The weather, or the time of day, for example, can make a big difference to playing conditions.

5. What is the most frequent decision taken for the winner of the toss? Batting or bowling? [1]
6. How many times did the winner of the toss also win the game? [1]
7. How likely are you to win the game if you win the toss? Show a plot indicating how this changes for all venues. [3]

3.7 Team scores

For this section you will need to calculate the total number of runs scored in each game. Keep in mind the following: each match has 1 or 2 innings (sometimes rain prevents the second team from batting), each innings consists of a maximum of 20 overs, and each over has up to 6 *legal* deliveries (no-balls and wides do not count). Each delivery will have 0 or more runs scored off it, and 0 or 1 wickets taken (dismissals). You will need to aggregate this data so that you can calculate the total number of runs scored per innings, and how many wickets were lost, and how many overs were completed. It might be useful to store this aggregated match data as a separate dataframe.

1. Calculate and plot the total number of runs scored per match. [2]
2. Next, plot the average number of runs scored per match in each season. Is there any noticeable trend? [2]
3. The higher the score in the 1st innings, the more likely this team is to win. Look at how frequently (ie what percentage of games are won) a game is won when the 1st innings total is above 190. [2]

Part of T20 cricket is that it is meant to be exciting. People want to see the players hit the ball a long way, scoring lots of runs, taking risks. If there are particular venues where more sixes are being hit, or more runs scored, then the organisers might want to schedule more games in this location, since they can draw bigger crowds.

4. How many 6s were hit in each season, in total? [1]
5. How many 6s were hit on average per game for each season? Are there more 6s being hit in the later seasons? [3]
6. Look at the average number of 6s per game at each venue. Are there specific venues that see more 6s being hit on average? Give some plausible explanations. [2]

3.8 Batting

Finding the best batsman for your team is not as simple as taking the player with the most runs, or the player that was paid the most in the previous year, or even the player that performed best for their national team. The conditions in different cities (weather, altitude) affect the way that the ball bounces, and how far the ball travels (air resistance), and this can suit some players, whilst be difficult for others. These aspects all

need to be considered when choosing which players you want in your team.

There are a few key metrics that batsmen get rated on. The most obvious one is how many runs they score on average, but as important is how *quickly* they score those runs. Additionally, there is the entertainment value that a player provides - how many fours and sixes they hit.

Batting terminology:

- Average score: sum all runs scored, and divide by the number of times they were dismissed. This is the average number of runs a player scores before being dismissed. A decent average in T20 would be 30 and above.
- Strike rate: sum all runs scored and divide by total number of legal deliveries faced and multiply by 100. This gives the expected number of runs a player will score if they face 100 deliveries. In T20 a player should have a strike rate of over 100.

In order to do a proper analysis of the players, it is useful to look at their performances per match, or over all time. To do this, you will need to aggregate the delivery dataset as follows:

1. For each player, calculate the following for each match where they batted (ie, aggregate over the match level):
 - total number of runs in each game
 - total number of sixes and fours in each game
 - total number of deliveries faced [5]
2. From this, create another dataset which contains the following for each player (ie aggregate over all matches):
 - total runs
 - total deliveries faced
 - total number of fours and sixes
 - total number of times dismissed
 - strike rate
 - average [5]

From the above datasets, we can start to look for trends in player performance.

3. Compute the total number of runs for all players over all time. Take the top 5 and show a bar plot with their names on the x-axis and the total runs on the y-axis. [2]
4. For the top run-scorer, plot all of their individual scores over time - have they improved, remained steady, or deteriorated? [2]
5. Plot the distribution of all averages for all batsmen. What percentage of batsman score an average of 30 or above, given a minimum of 500 runs? [2]

6. Which players have hit the most sixes? How does this compare against the list of players that have the highest average number of sixes per game. [3]
7. For all batsmen, is there an identifiable correlation between runs scored, average, strike rate, and number of sixes? You can look at scatter plots, heatmaps, etc. [4]

3.9 Bowlers

Bowling terminology:

- *Economy rate*: sum up all runs scored off a bowler, and divide this by the number of deliveries that the bowler delivered, then multiply by 6. This metric shows how many runs are scored, on average, off a single over bowled by this player.
- *Average*: sum up all runs scored against this bowler and divide this by the sum of all wickets taken by this bowler. This tells you how many runs are scored off a bowler, on average, between every wicket they take.
- *Wickets (or dismissals)*: When a batsman gets dismissed (when they go out), the credit for this does not always go to the bowler. If the dismissal was a run-out, retired, or obstructed field, then the bowler does not get to claim this dismissal as their own.

As above for the batsman, we also want to aggregate the data for the bowlers.

1. For each player, calculate the following for each match where they bowled (ie, aggregate over the match level):
 - total number of wickets in the game
 - total number of deliveries bowled
 - total number of runs scored off their bowling [5]
2. From this, create a dataset which contains the following for each player (ie aggregate over all matches):
 - total wickets
 - total deliveries bowled
 - total runs scored off their bowling
 - bowling economy rate
 - bowling average [5]

Using the above datasets, we can look at bowler performances in more detail.

3. Plot the distribution of wickets per player and then identify the top 5 bowlers with the most wickets. [2]
4. From all players that have bowled at least 20 overs (120 deliveries), identify the top 5 bowlers with the lowest economy rate from this selection. [1]
5. From all players that have bowled at least 20 overs (120 deliveries), identify the top 5 bowlers with the lowest average from this selection. [1]

6. Look at the correlations between wickets taken, bowling average, and economy rate. Are there any obvious correlations? [4]

Certain bowlers tend to perform well against particular batsmen, dismissing them more frequently than other bowlers. We can do a pairwise comparison between all bowlers and batsmen to identify this.

7. For the top bowler, identify if there are particular batsman that they dismiss more frequently than others. [2]
8. Identify the amount of runs scored against this bowler by any batsman. Show the top 5 results. [3]

We have now looked at individual metrics for all these players. A less quantifiable metric is the player of the match award. This is voted on and is somewhat subjective, but it does still give an indication of how much of an impact a player had in a game.

9. Do a count of how frequently each player has received a player of the match award. Do these match what you would expect for the top 5 batsmen or bowlers? (Top 5 based on any of the metrics). [2]

There is a lot more that can be done with this dataset. Looking at the umpires and how they make decisions on leg before wickets, for example. Knowing which players perform poorly against specific teams, or in which types of conditions can also provide insights. Which venues are preferred by which players or teams. Looking at how the top paid players actually performed (<https://www.iplt20.com/auction>) and if they were worth the salary paid to them. You could also obtain viewership information, and ticket sales per game. Unfortunately this was given to you as an exam, and there is limited time, but I hope you will find this interesting enough to keep on playing with this dataset!